



AURA · PHASE 01 — Project Foundation, Catalog & Wedding Project Ingest · Full Build Workspace



Phase 01 done means: point AURA at a folder of 4,000 RAW files, walk away, kill the process at any moment, relaunch, and it resumes to a complete, correct, deduplicated catalog with zero duplicate rows and every failure coded and quarantined. No AI yet. Just an unbreakable foundation.

1. Mission

Everything in phases 02–30 writes into the catalog this phase creates and depends on the error, ID and job contracts this phase freezes. If these are wrong, thirty phases inherit the mistake. Nothing here is glamorous and all of it is load-bearing.

Owners: CTO TLC SRC SFE DEVOPS · **Estimate:** 2 weeks · **Epic:** E1

Unlocks: PHASE-02 (RAW decode needs `photo` rows and the preview cache layout), PHASE-03 (model manager needs the paths and error model), PHASE-04 (cloud gateway needs `ProjectConsent`), PHASE-28 (autopilot needs the job graph).

2. Scope

In scope

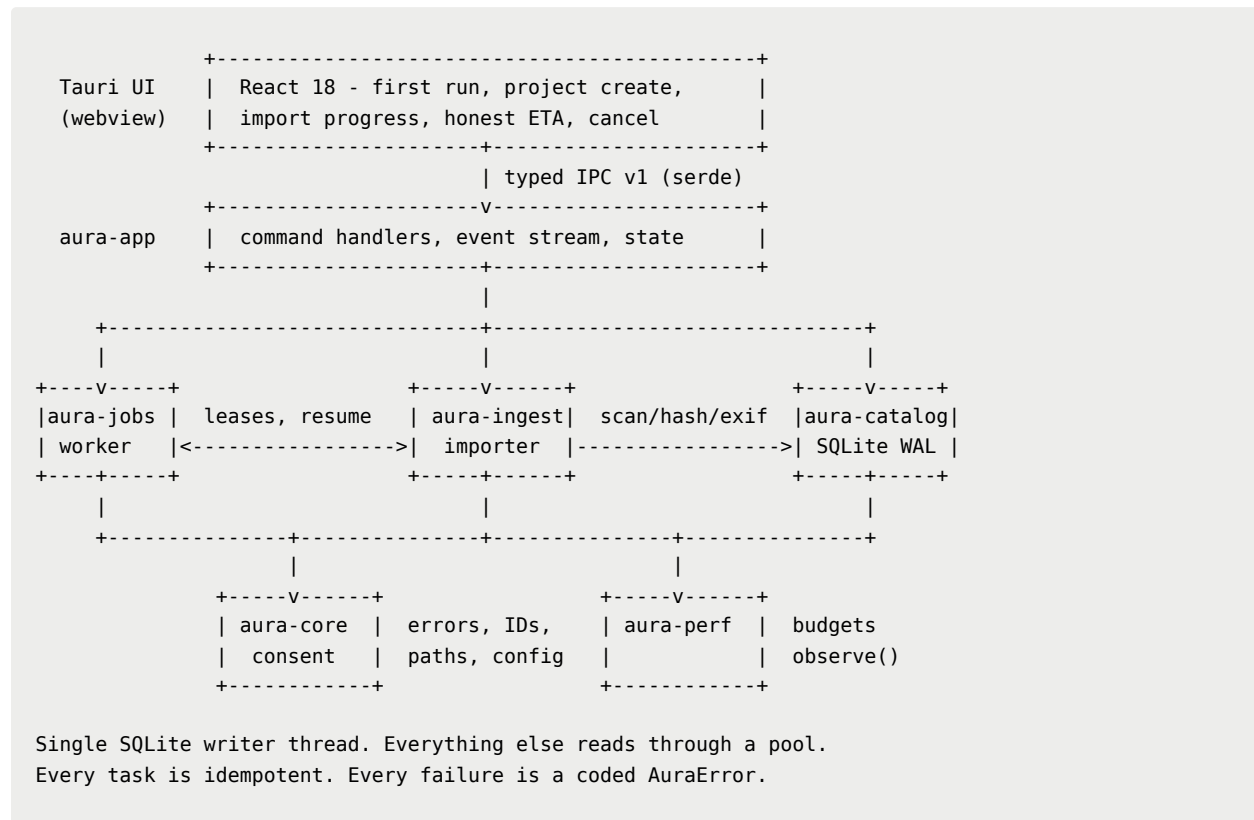
- Cargo workspace, 9 crates, toolchain pinning, lint and banned-pattern gates
- The frozen error taxonomy with the `errors.toml` registry and runbook gate
- Typed IDs, path resolution, app config, `ProjectConsent`
- Catalog schema v1 with migrations, verified backups, integrity checks, single-writer enforcement
- Cloud-sync folder refusal and dual-instance locking

- Ingest: recursive scan, format classification, BLAKE3 content hashing, dedupe, EXIF extraction, sidecar pairing, idempotent transactional import, quarantine
- Job graph skeleton with leases, heartbeats and resume — enough to run ingest tasks
- Tauri shell, typed versioned IPC, first-run project creation, honest progress, cancel
- Fixtures, unit / integration / property / chaos tests, CI stage-1 and stage-2 lanes

Out of scope — do not let the agent drift into these

- RAW decoding or preview generation (PHASE-02). Phase 01 reads **embedded JPEG previews only** for thumbnails, and only if trivially available.
- Any model, any inference, any network call.
- The virtualised grid beyond a stub that proves 60 fps with 4,000 placeholder cells.
- Editing, culling, ratings, XMP writing.

3. Architecture of this phase



4. Repo layout created by this phase

```

aura/
  Cargo.toml                workspace root
  rust-toolchain.toml        pinned 1.83.0
  clippy.toml rustfmt.toml  deny.toml
  .github/workflows/ci.yml
  scripts/check-banned.sh
  contracts.lock
  crates/
    aura-core/              errors, ids, paths, config, consent
    aura-catalog/           schema, migrations, pool, single writer, backup
    aura-ingest/            scan, classify, hash, exif, import, quarantine
    aura-jobs/              graph contract, store, leases, worker
    aura-perf/              budget observe/assert
    aura-app/               Tauri commands, events, ipc contract
    aura-cli/               headless import for CI and support
    xtask/                  contracts check, bench, fixtures
  ui/                       React 18 + TS + Vite
  migrations/0001_init.sql
  tests/fixtures/           generated corpora
  docs/runbooks/           one per error code
  docs/adr/

```

5. Contracts frozen by this phase

Once these land in `contracts.lock`, changing any of them requires an ADR with a grep blast radius, per `13-CONTRACT-FREEZE-AND-ADR.md`.

Contract	File	Consumed by
<code>AuraError</code> , <code>Severity</code> , <code>Recovery</code>	<code>aura-core/src/contract/error.rs</code>	Every crate, every phase
Typed IDs	<code>aura-core/src/contract/ids.rs</code>	Every crate
<code>ProjectConsent</code>	<code>aura-core/src/contract/consent.rs</code>	PHASE-04, 07, 13, 27
Catalog schema v1	<code>migrations/0001_init.sql</code>	Every phase that persists anything
<code>ImportPlan</code> , <code>ImportedFile</code>	<code>aura-ingest/src/contract/ingest.rs</code>	PHASE-02, 30
<code>TaskId</code> , <code>TaskState</code> , <code>Lease</code> , <code>CostHint</code>	<code>aura-jobs/src/contract/graph.rs</code>	PHASE-02–30, especially 28
IPC v1	<code>aura-app/src/ipc/contract.rs</code> • <code>ui/src/ipc/types.ts</code>	Every UI phase

6. Error codes registered by this phase

Code	Meaning	Severity	Recovery
<code>AURA-IO-1001</code>	Source path does not exist	RunBlocking	AskUser

Code	Meaning	Severity	Recovery
AURA-IO-1002	Permission denied	RunBlocking	AskUser
AURA-IO-1003	Volume disconnected mid-run	RunBlocking	Retry
AURA-IO-1004	Disk full	RunBlocking	Halt
AURA-IO-1005	Path exceeds OS limit	ItemFailed	Quarantine
AURA-IO-1006	File unreadable or truncated	ItemFailed	Quarantine
AURA-IO-1007	File changed during read	ItemFailed	Retry
AURA-IO-1008	Catalog location is cloud-synced	RunBlocking	AskUser
AURA-IO-1010	Zero-byte file	ItemFailed	Quarantine
AURA-DB-3001	Catalog open failed	Fatal	AskUser
AURA-DB-3002	Migration failed, backup restored	Fatal	Halt
AURA-DB-3003	Integrity check failed	Fatal	AskUser
AURA-DB-3004	Catalog schema newer than app	Fatal	AskUser
AURA-DB-3005	Second instance holds the write lock	RunBlocking	AskUser
AURA-DB-3006	Pre-migration backup failed	Fatal	Halt
AURA-JOB-7001	Task lease expired, reclaimed	Warning	Retry
AURA-JOB-7002	Task failed after max attempts	ItemFailed	Quarantine
AURA-JOB-7003	Unknown or cyclic task dependency	Fatal	Halt
AURA-JOB-7004	Lease lost while working, result discarded	Warning	Retry
AURA-JOB-7005	Resume found inconsistent state	RunBlocking	Halt

Every code above needs `docs/runbooks/<code>.md` before CI goes green.

7. Budgets for this phase, measured on T-Typ

Measurement	Target	Ceiling
Scan + hash + EXIF + insert, 4,000 files / 220 GB, USB 3.2 SSD	5 min	9 min
Sustained hash throughput	700 MB/s	≥ 400 MB/s
Catalog open, 200,000 photo rows	250 ms	600 ms
Peak RSS during import	700 MB	1.2 GB
Resume after kill -9	2 s	5 s
Cold app launch to interactive window	1.2 s	2.5 s
Cancel acknowledged	200 ms	500 ms
Migration on 200,000 rows	1 s	5 s

8. Task table — every job has one owner

#	Task	Role	Deliverable	Est.
T1	Workspace, crates, pinned toolchain, lints, banned-pattern script	DEVOPS	<code>Cargo.toml</code> , <code>clippy.toml</code> , <code>scripts/check-banned.sh</code>	0.5 d
T2	Error taxonomy, <code>errors.toml</code> , runbook gate, <code>xtask</code> <code>contracts</code>	TLC	<code>aura-core/src/contract/error.rs</code>	1 d
T3	Typed IDs, path resolver, config, consent contract	TLC	<code>ids.rs</code> , <code>paths.rs</code> , <code>config.rs</code> , <code>consent.rs</code>	1 d
T4	Catalog schema v1 DDL with indices and constraints	SRC	<code>migrations/0001_init.sql</code>	1 d
T5	Migration runner, verified backup, integrity check, version guard	SRC	<code>aura-catalog/src/migrate.rs</code>	1 d
T6	Single-writer actor, read pool, cloud-sync refusal, instance lock	SRC	<code>aura-catalog/src/db.rs</code> , <code>guard.rs</code>	1.5 d
T7	Recursive scan, classification, long-path and Unicode handling	SRC	<code>aura-ingest/src/scan.rs</code>	1 d
T8	BLAKE3 hashing, dedupe, sidecar pairing, RAW+JPEG pairing	SRC	<code>aura-ingest/src/hash.rs</code> , <code>pair.rs</code>	1 d

#	Task	Role	Deliverable	Est.
T9	EXIF extraction with hostile-input hardening	SRC • COL	aura-ingest/src/exif.rs	1 d
T10	Idempotent transactional import, quarantine, import report	SRC	aura-ingest/src/import.rs	1.5 d
T11	Job graph contract, store, one-statement claiming, dependency promotion	TLC	aura-jobs/src/contract/graph.rs , store.rs	1.5 d
T12	Lease reclaim, startup recovery, worker loop, eight-kill-point chaos test	TLC	aura-jobs/src/reclaim.rs , worker.rs	1.5 d
T13	Tauri shell, typed IPC v1, event stream, cancel	SFE	aura-app/src/ipc/* , ui/src/ipc/types.ts	1.5 d
T14	First-run flow, project creation, consent screen, four refusals	SFE • UX	ui/src/screens/*	2 d
T15	Honest progress component and grid stub at 60 fps	MFE	ui/src/components/*	1 d
T16	Fixture generator and the four test corpora	QAL	xtask fixtures , tests/fixtures/	1 d

#	Task	Role	Deliverable	Est.
T17	Unit, integration, property and chaos suites	QAL	tests/**	2 d
T18	CI lanes, local gate script, budget assertions, flake policy, artefacts	DEVOPS	.github/workflows/ci.yml , perf/budgets.toml , aura-perf/src/budget.rs	1.5 d
T19	Runbooks for all 19 codes plus the four operational runbooks	DOC	docs/runbooks/*.md	1 d
T20	Two-hat review, DoD audit, exit report	EM • QAL • CTO	docs/phase-exit/PHASE-01.md	1 d

9. Test plan

Level	What it proves
Unit	Classification, hashing, EXIF parsing, path normalisation, ID round-trips, error mapping
Property	Import is idempotent for any file set; scan order never changes the resulting catalog
Integration	Full import of the 4,000-file fixture; migration chain; backup and restore; dual-instance lock
Chaos	kill -9 at 8 points; volume yanked mid-import; disk filled at 90 %; clock jumped; hostile filenames
Performance	Every budget in section 7, asserted not merely logged
Security	Poisoned-path fixture: no C2 data in any log, event, or bundle

10. Acceptance criteria — phase 01 is done when all are true

- ☐ `cargo build --workspace --all-targets` clean; Clippy clean with `-D warnings`
- ☐ `scripts/check-banned.sh` finds zero `unwrap` , `expect` , `panic!` , `todo!` outside tests and `main`
- ☐ All 7 contracts in section 5 exist verbatim and are listed in `contracts.lock`

- ☐ All 19 codes in section 6 are in `errors.toml` and each has a runbook that `cargo xtask runbooks --check` accepts
- ☐ 4,000-file fixture imports fully; re-running the identical import inserts **zero** new rows
- ☐ kill -9 at all 8 chaos points resumes to the identical final catalog digest
- ☐ Two AURA instances on one catalog: the second refuses with `AURA-DB-3005`
- ☐ A catalog placed in a Dropbox or OneDrive folder is refused with `AURA-IO-1008` and an actionable message
- ☐ Every budget in section 7 measured within ceiling and pasted in the exit report
- ☐ Poisoned-path test green; no filename appears in any log or event
- ☐ Cancel during import stops within 500 ms and keeps completed work
- ☐ UI reaches interactive in under 2.5 s and scrolls 4,000 cells at 60 fps
- ☐ Two-hat review completed with recorded findings; `PHASE-01` exit report signed by `CT0`

11. Claude Code kickoff prompt

Read `CLAUDE.md`, `AGENT-TEAM.md`, `11-ERROR-TAXONOMY.md`, `12-DETERMINISM-CONTRACT.md`, `13-CONTRACT-FREEZE-AND-ADR.md`, `15-CONCURRENCY-AND-JOB-GRAPH.md`, `17-RESOURCE-BUDGETS.md`, `21-DEFINITION-OF-DONE.md` in full, then this page and every deliverable page beneath it.

Confirm back to me: the 9 product invariants, the 5 reliability invariants, the banned patterns, and the 7 contracts this phase freezes. Then implement T1 only, wearing the `DEVOPS` hat, and stop. Do not start T2 until I approve T1.

Deliverables

Ten pages. Every frozen contract, every SQL statement, every test and every command needed to build PHASE-01 with nothing left to invent. Work them in order; page 10 is the one you actually run from.

1.  P01-01 · T1 Repo Scaffold, Toolchain & Banned-Pattern Gate — DEVOPS
 - workspace manifest, pinned 1.83.0 toolchain, clippy denials, `check-banned.sh`, BLAKE3 contract digests into `contracts.lock`
2.  P01-02 · T2-T3 aura-core — Error Taxonomy, Typed IDs, Paths, Clock & Consent
 - `AuraError`, the code registry, typed ids that cannot be mixed, injectable clock, cloud-sync refusal, consent that starts denied

3. — the full `0001_init.sql`, every pragma with its reason, the single-writer actor, advisory lock, verified backup before any migration
4. — deterministic scan order, streaming BLAKE3, RAW+JPEG+sidecar pairing, degrade-never-fail EXIF, import that is idempotent by construction
5. — the state machine as the single source of truth, one-statement claiming, startup reclaim, and the eight-kill-point convergence test
6.  P01-06 · T13-T15 Tauri Shell, Typed IPC v1, First Run & Honest Progress
— the IPC contract mirrored in Rust and TypeScript, four first-run refusals with exact wording, progress that refuses to guess, 4,000-cell grid
7.  P01-07 · T16-T17 Fixture Generator & the Full Test Suite
— deterministic corpus generation, the 60-file hostile corpus, 14 property tests, 11 real chaos scenarios, the poisoned-path privacy test
8.  P01-08 · T18 CI Lanes, Local Gates & Budget Instrumentation
— `budgets.toml` as the single source of truth, budgets that fail builds, two-stage PR lane, nightly chaos and perf, flake hunt
9.  P01-09 · T19-T20 Runbooks, DoD Evidence Pack & Phase Exit Report
— 19 error-code runbooks plus 4 operational ones, the 28-row evidence table, the 15-step human walkthrough, the signed exit report
10.  P01-10 · Execution Script — T1 to T20, Day by Day, Prompt by Prompt
— the ten-day schedule, the session opener, twenty ordered prompts with stop points, and the six hard-stop conditions

How to use this workspace

-  Open page 10, paste the session opener, then paste the Day 1 prompt. When it stops and hands back evidence, tick the acceptance boxes on the matching page — not from memory, from the pasted output. Only then move to the next prompt. Anything marked **FROZEN** is copied verbatim, comments included; its BLAKE3 digest is checked in CI, so an "improvement" fails the build.

 P01-01 · T1 Repo Scaffold, Toolchain & Banned-Pattern Gate — DEVOPS

 P01-02 · T2-T3 aura-core — Error Taxonomy, Typed IDs, Paths, Clock & Consent

 P01-03 · T4-T6 aura-catalog — Schema v1, Migrations, Single Writer & Locks

-  P01-04 · T7–T10 aura-ingest — Scan, Hash, Pair, EXIF & Idempotent Import
-  P01-05 · T11–T12 aura-jobs — Job Graph, Leases, Heartbeats & Crash Resume
-  P01-06 · T13–T15 Tauri Shell, Typed IPC v1, First Run & Honest Progress
-  P01-07 · T16–T17 Fixture Generator & the Full Test Suite
-  P01-08 · T18 CI Lanes, Local Gates & Budget Instrumentation
-  P01-09 · T19–T20 Runbooks, DoD Evidence Pack & Phase Exit Report
-  P01-10 · Execution Script — T1 to T20, Day by Day, Prompt by Prompt